

Question Number	Answer	Additional Guidance	Mark
3(a)(i)	area of zone {with no bacteria / of inhibition}		(1)

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	to allow (chemicals in the extract) to diffuse into agar / to stop growth of bacteria (1)		(1)

Question Number	Answer	Additional Guidance	Mark
3(a)(iii)	An description that includes the following points: - measure {diameter / radius} of zone (1) - apply $\text{area} = \pi r^2$ (1) OR - place dish on {gridded / graph} paper (1) - count squares (1)	ACCEPT $\pi(d/2)^2$ ACCEPT $r = d \div 2$ ACCEPT cut out and weigh	(2)

Question Number	Answer	Additional Guidance	Mark
3(a)(iv)	An answer that includes the following points: • filter paper discs qualified (1) • (soaked in) {solvent / water} (1)	e.g. size, paper type, thickness, same as ones used	(2)

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	An answer showing the following steps: • difference calculated (1) • percentage calculated (1)	Correct answer with no working shown gains two marks • 196-111 or 85 (1) • $85 \div 111 = 76.57 / 76.6$ / $85 \div 196 = 43.4 / 43.37$ (1)	(2)

Question Number	Answer	Additional Guidance	Mark																		
3(b)(ii)	A graph showing the following features: • A bar chart (1) • L axes correctly labelled, and with units (mm²) (1) • P correct plotting (1) • K key or labels to show type A and type B bacteria results (1) • S suitable linear scale for area starting at zero with no interruption (1)	area of zone of inhibition / mm² A B <table><thead><tr><th>Species</th><th>Type A (mm²)</th><th>Type B (mm²)</th></tr></thead><tbody><tr><td>clove</td><td>200</td><td>110</td></tr><tr><td>cumin</td><td>70</td><td>0</td></tr><tr><td>ginger</td><td>190</td><td>0</td></tr><tr><td>pomegranate</td><td>270</td><td>160</td></tr><tr><td>thyme</td><td>240</td><td>0</td></tr></tbody></table>	Species	Type A (mm ²)	Type B (mm ²)	clove	200	110	cumin	70	0	ginger	190	0	pomegranate	270	160	thyme	240	0	
Species	Type A (mm ²)	Type B (mm ²)																			
clove	200	110																			
cumin	70	0																			
ginger	190	0																			
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thyme	240	0																			

(5)

Question Number	Answer	Additional Guidance	Mark
3(b)(iii)	An answer that includes three of the following points: • cumin is the least effective extract to use (1) • pomegranate is the most effective extract to use (1) • cumin, thyme, and ginger cannot be used to preserve food against type B (1) • type B is more difficult to control (1)	ACCEPT against Type A ACCEPT pomegranate and clove can be used against B ACCEPT converse / A inhibited by all	(3)

Question Number	Answer	Additional Guidance	Mark
3(c)	An answer that includes two of the following points: • {solubility / concentration / volume} of extract / solvent used (1) • size of molecules in the extract (1) • rate of diffusion of extract into agar (1)	ACCEPT antimicrobial	(2)